

Decent Coaching Classes

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Answer Key — Basic Geometrical Concepts & Understanding Elementary Shapes

Class VI — Mathematics (NCERT) — Chapters 2 & 3 | Maximum Marks: 30

Section A — Multiple Choice Questions

- A1. (a) 0 [1]
A2. (b) polygon [1]
A3. (c) sides [1]
A4. (b) straight angle [1]
A5. (c) 90° [1]
A6. (c) Equilateral triangle [1]

Section B — Fill in the Blanks / Very Short Answer

- A7. one (1) [1]
A8. diagonal [1]
A9. Hexagon [1]
A10. Square [1]

Section C — Short Answer Questions

- A11. A line segment is a fixed part of a line with two definite end points and a fixed, measurable length. A line extends indefinitely in both directions, has no end points, and its length cannot be measured. [2]
A12. (a) A curve shaped like the letter 'C' does not start and end at the same point, so it is an Open curve. (b) A curve shaped like the letter 'O' starts and ends at the same point, so it is a Closed curve. [2]
A13. (a) 3 sides — Triangle. (b) 4 sides — Quadrilateral. (c) 8 sides — Octagon. [2]
A14. (a) $\triangle ABC$ has $AB = 5$ cm, $BC = 6$ cm, $CA = 7$ cm — all sides unequal, so it is a Scalene triangle. (b) $\triangle PQR$ has $PQ = PR = 4.5$ cm (two equal sides), so it is an Isosceles triangle. [2]
A15. (a) True — a square has all sides equal and all angles 90° , which satisfies the definition of a rectangle (opposite sides equal and parallel, all angles 90°), so every square is a rectangle. (b) False — a rectangle need not have all four sides equal, so every rectangle is not necessarily a square. [2]
A16. Turning clockwise from North to East is a quarter turn = 90° , which is equal to 1 right angle. [2]

Section D — Long Answer / Diagram-Based Questions

- A17. Steps of construction: (1) Draw a ray BA. (2) Place the centre of the protractor at B, with the 0° -line along BA. (3) Mark a point C on the paper against the 65° mark on the protractor. (4) Remove the protractor and draw ray BC. $\angle ABC$ so obtained = 65° . (Award marks for a correctly drawn angle close to 65° with all four steps stated.) [4]

A18. (a) A hexagon has 6 vertices; from any one vertex, diagonals can be drawn to the 3 non-adjacent vertices, giving 3 diagonals from that vertex (the two adjacent vertices are joined by sides, not diagonals). A correctly labelled rough hexagon with these 3 diagonals drawn from one vertex should be awarded full marks. (b) Any simple closed curve that is not made up entirely of line segments — for example, a circle or an oval/wavy closed shape — is acceptable, since a polygon must be bounded only by straight line segments. [4]

— End of Answer Key —